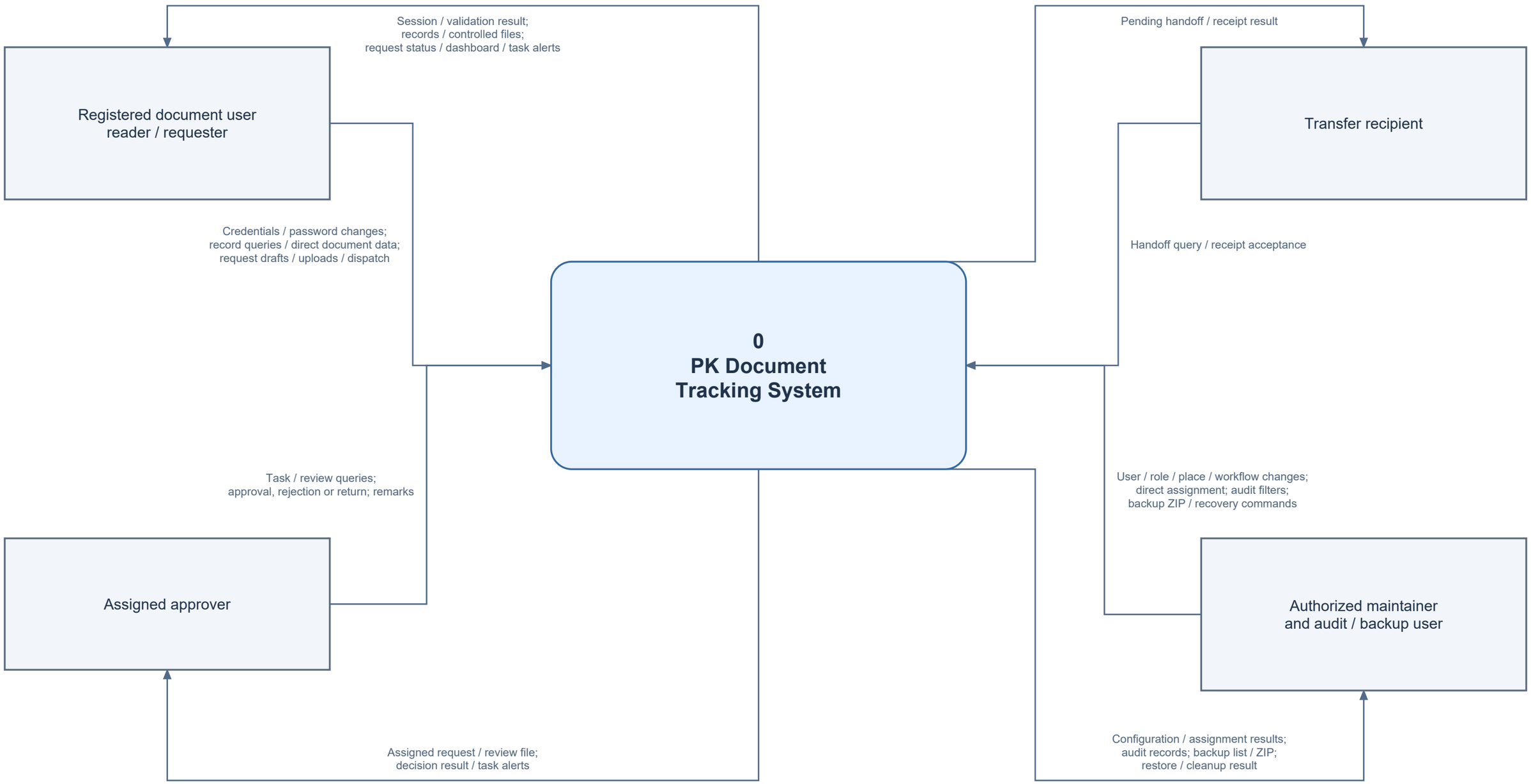
Context diagram

PK Document Tracking System - implemented data exchanges

External participant • Process / activity → Data flow



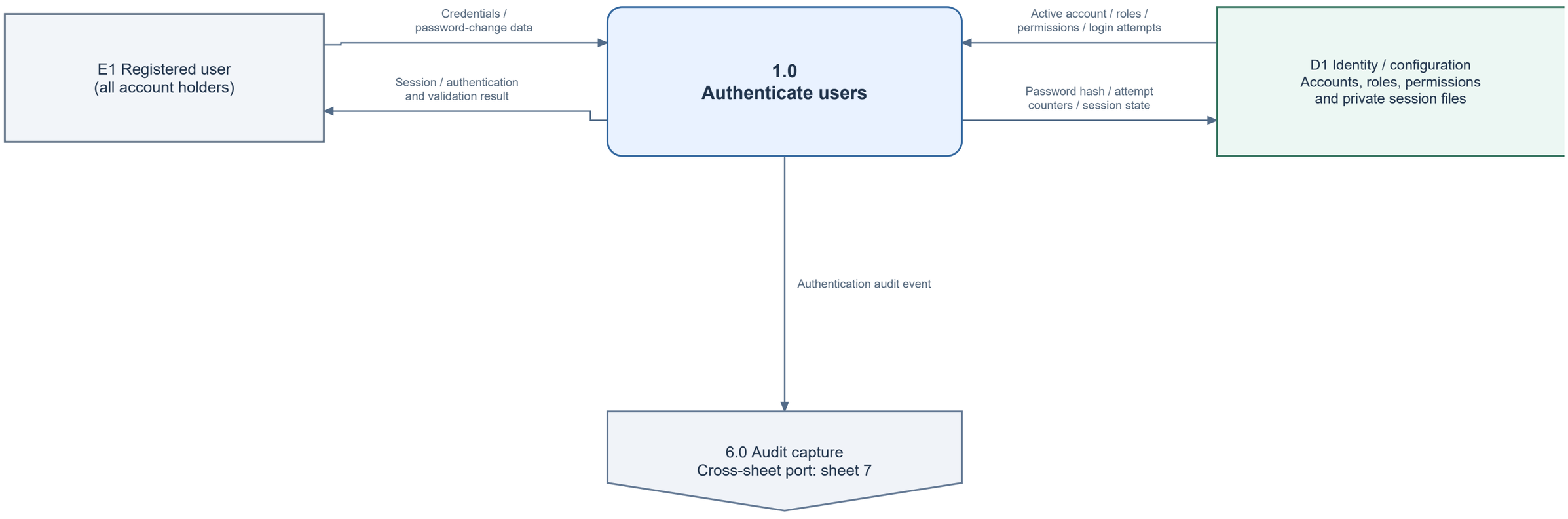
- Actors are functional responsibilities and may be performed by the same account. Role permissions determine available actions; an approver is selected by the published workflow.
- Only information about physical document dispatch and receipt crosses the software boundary. The paper itself is not a software data flow.
- All account holders authenticate as registered users. No email, SMS or external document cloud is evidenced in the operational code.
- Level 0 is presented as six coordinated subsystem sheets (2-7) using the same process and store identifiers. Cross-sheet ports identify shared audit capture.

Source evidence: application/config/routes.php:26-232 | application/controllers/Requests.php:16-221 | application/controllers/Administration.php:40-448 | application/controllers/Files.php:22-239 | application/controllers/Backups.php:6-155

Level 0 / 1.0 - authenticate users

Account credentials become a checked server identity and session; all actors use this registered-user role.

External participant • Process / activity • Data store • Cross-sheet reference → Data flow



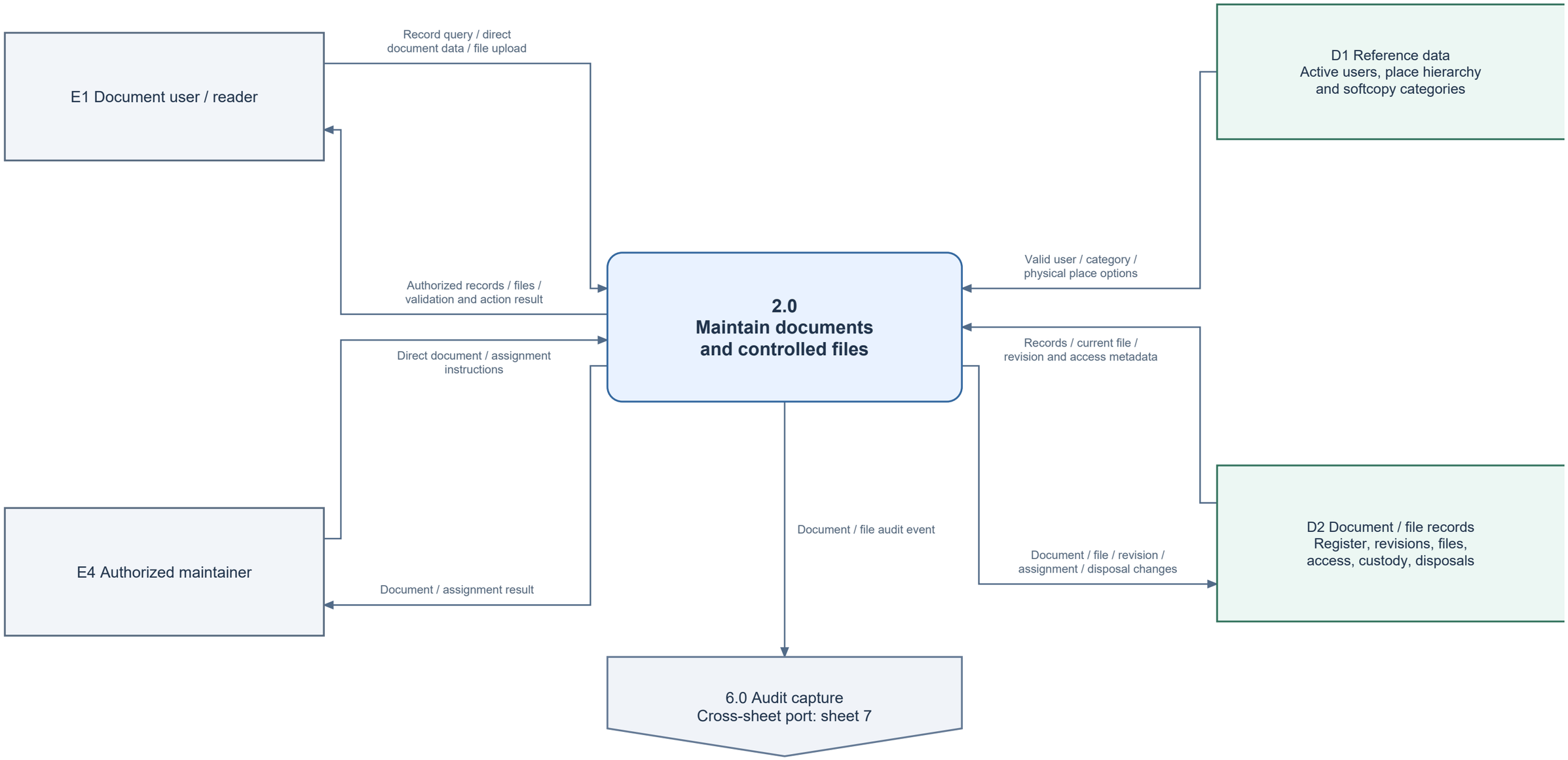
- Identity comes from active users and active assigned roles. The Administrator role grants wildcard permissions; other accounts receive the union of their active role grants.
- Password verification, first-login password setup and session_version checks are implemented. No external identity provider or email password-recovery flow is depicted.
- D1-D4 are logical store groups; see the physical store catalog on sheet 8. A cross-sheet port refers to the numbered process on its named sheet.

Source evidence: application/controllers/Auth.php:24-75,95-138 | application/models/Identity_model.php:6-40 | application/models/Permission_model.php:23-49 | application/config/config.php:21-24

Level 0 / 2.0 - maintain documents and files

Authorized direct changes, scoped browsing and controlled file delivery use the same document register.

External participant • Process / activity • Data store • Cross-sheet reference → Data flow



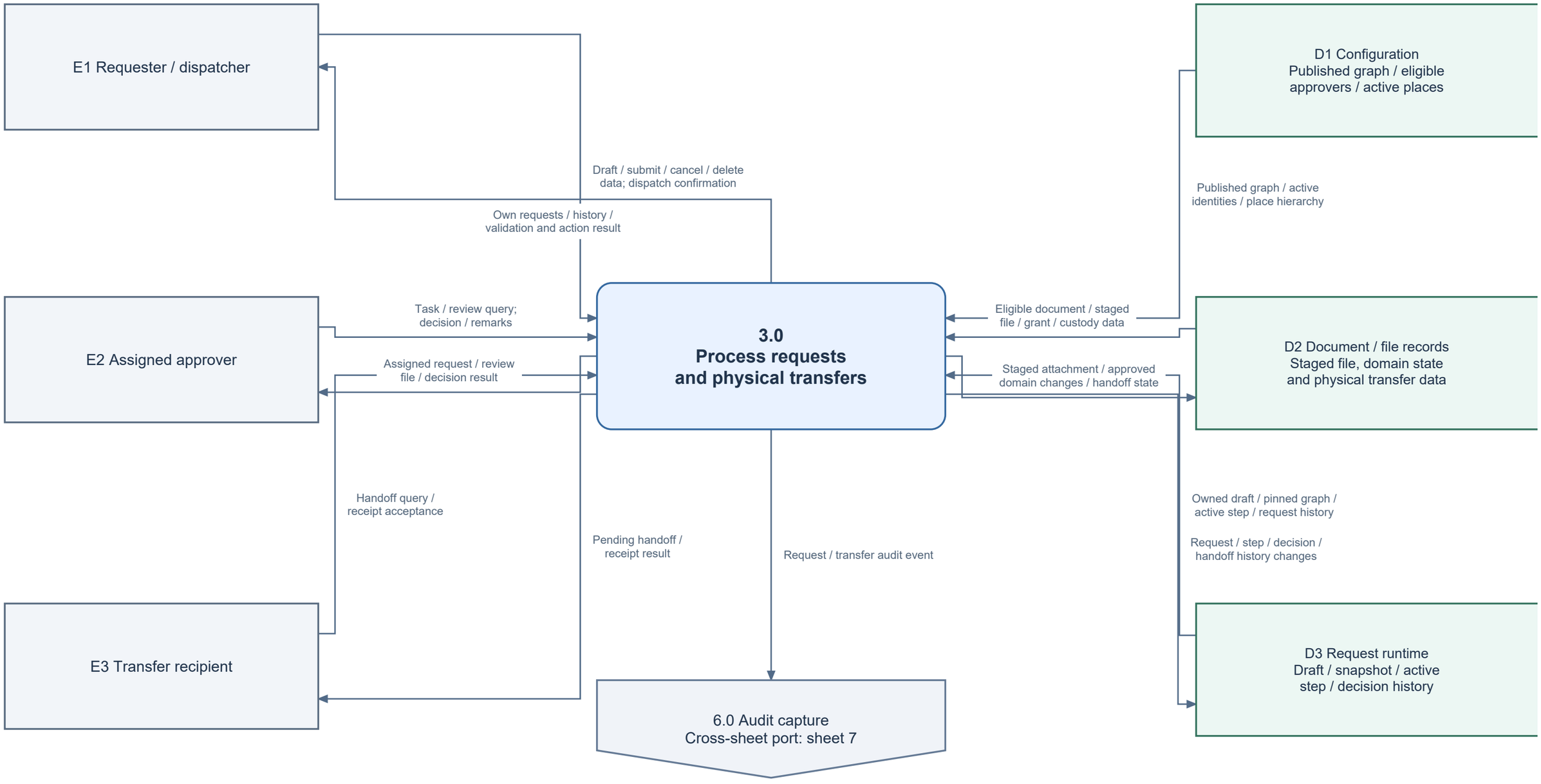
- Direct actions require module permissions and authorized document scope. Controlled upload bytes are private; approved file access also checks document permissions and scope.
- Softcopy assignment uses assignments.softcopy_id. Hardcopy assignment changes hardcopy_documents.holder_id; no synthetic hardcopy_id column exists in assignments.
- Office/WPS review is an optional local application handoff using a signed content URL. File review, download and printing remain protected application routes.
- D1-D4 are logical store groups; see the physical store catalog on sheet 8. A cross-sheet port refers to the numbered process on its named sheet.

Source evidence: application/controllers/Documents.php:17-194 | application/controllers/Files.php:22-239 | application/controllers/Administration.php:130-338 | application/services/documents/document_access_service.php:18-85 | application/services/files/file_service.php:171-228

Level 0 / 3.0 - process requests and transfers

Draft payloads and decisions become pinned approval records, applied domain changes and recorded custody.

External participant • Process / activity • Data store • Cross-sheet reference → Data flow



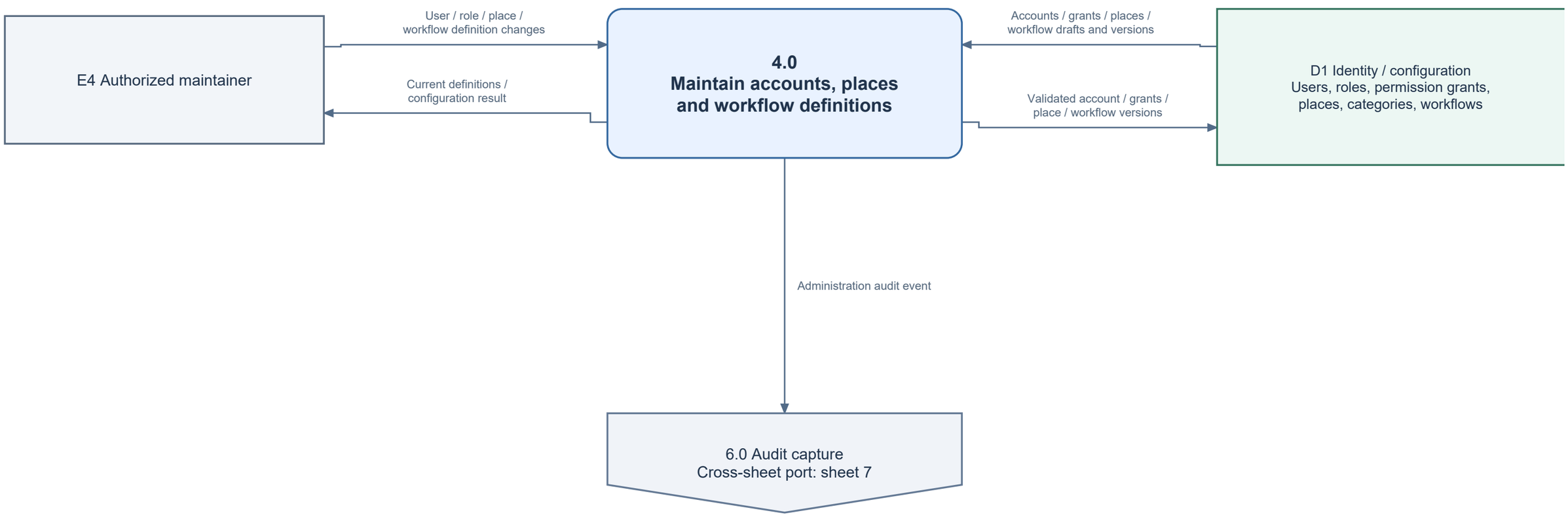
- Request types: softcopy_create, softcopy_revise, softcopy_cancel, hardcopy_create, hardcopy_update, transfer, disposal, access and assignment. Draft deletion is limited to the owner's cancelled, unsubmitted request.
- Submission pins a published workflow graph/version and resolves user, role, requester or requester_leader assignments. Approve/reject/return updates the active step and history; only final approval applies the domain proposal atomically.
- An approved transfer creates for_transfer data. Only the transfer requester dispatches it; only the named recipient accepts it. Acceptance updates holder/location, revokes the previous holder's personal grant and completes the request.
- D1-D4 are logical store groups; see the physical store catalog on sheet 8. A cross-sheet port refers to the numbered process on its named sheet.

Source evidence: application/models/Request_model.php:6-40 | application/services/request/request_service.php:116-317,393-543,565-839,849-906 | application/services/transfer/transfer_service.php:13-143

Level 0 / 4.0 - maintain system definitions

Validated administrative data defines account access, document locations and published approval routes.

External participant • Process / activity • Data store • Cross-sheet reference → Data flow



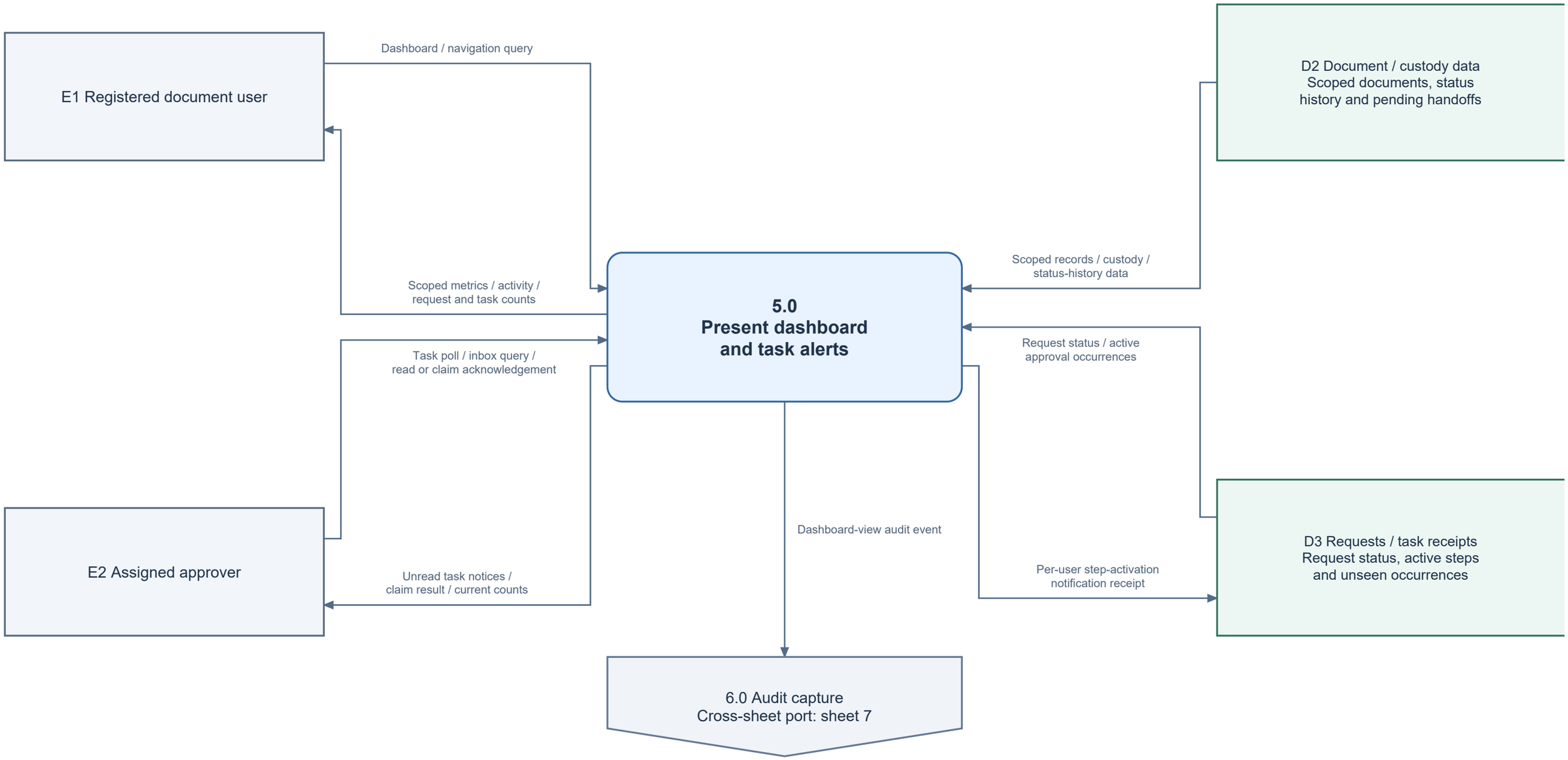
- A user may hold multiple active roles. User/role writes enforce delegation rules and invalidate sessions when necessary. Role deactivation/removal checks existing usage.
- Hardcopy place data comprises areas, specifics, assets and locations; softcopy categories form a parent-child hierarchy. Published workflows are versioned; an editable clone is a separate draft.
- D1-D4 are logical store groups; see the physical store catalog on sheet 8. A cross-sheet port refers to the numbered process on its named sheet.

Source evidence: application/controllers/Administration.php:340-448 | application/controllers/Places.php:31-88 | application/services/administration/administration_service.php:42-277 | application/services/places/place_service.php:12-200 | application/services/workflow/workflow_builder_service.php:87-201

Level 0 / 5.0 - present dashboard and task alerts

Scoped record counts and active approval assignments become dashboard metrics and in-application notices.

External participant • Process / activity • Data store • Cross-sheet reference → Data flow



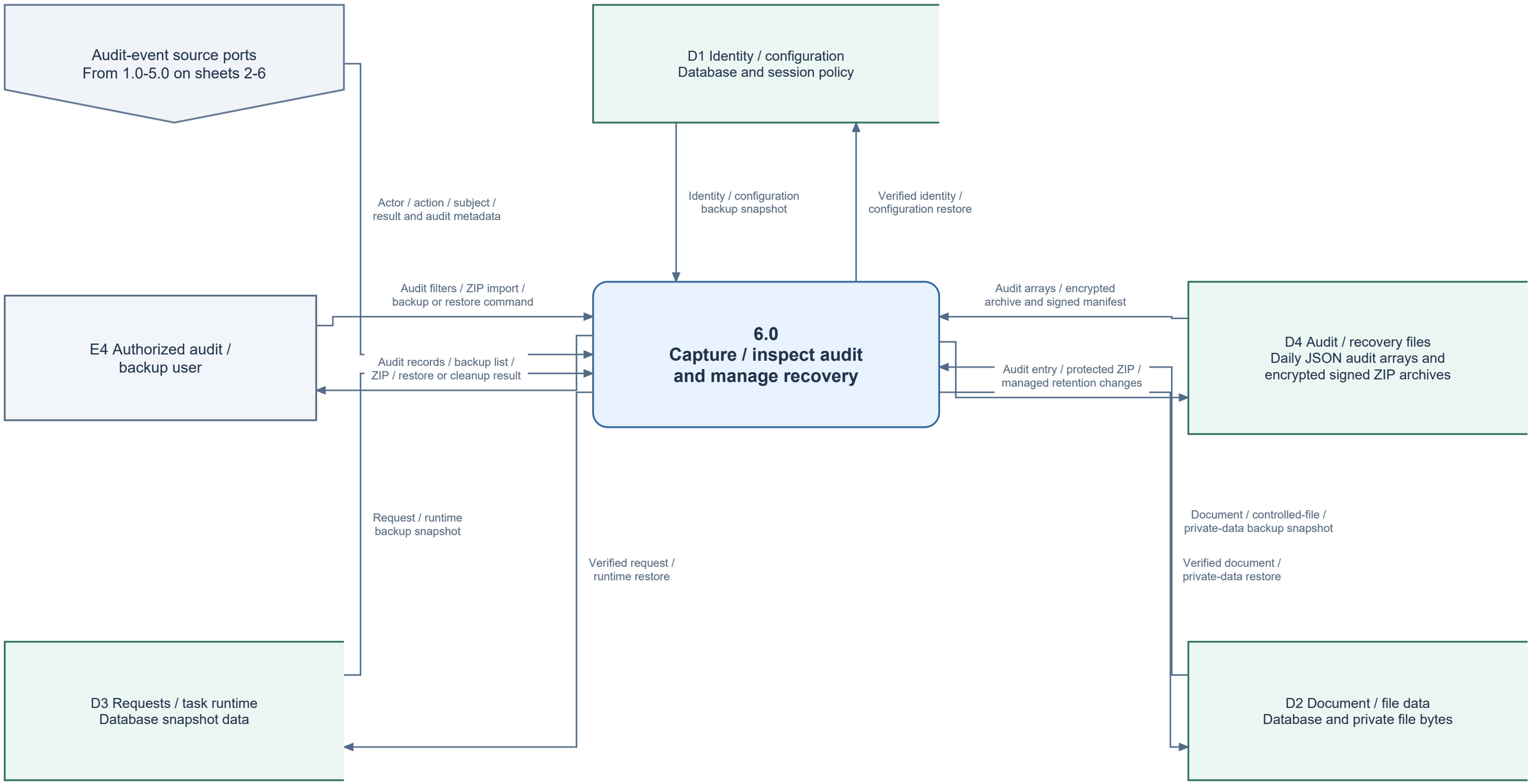
- Current task notices derive requests + workflow_steps + task_notification_receipts. An occurrence is keyed by step_id and step activation/version; atomic claims prevent repeat display across tabs/devices.
- The schema contains notifications, but the current inbox/pending implementation does not read it. No email, SMS or push-provider delivery is evidenced; notices are application HTTP/JSON responses.
- D1-D4 are logical store groups; see the physical store catalog on sheet 8. A cross-sheet port refers to the numbered process on its named sheet.

Source evidence: application/models/Dashboard_model.php:7-84 | application/models/Notification_model.php:26-225 | application/controllers/Notifications.php:25-91 | application/core/MY_Controller.php:138-157

Level 0 / 6.0 - capture audit and manage recovery

Operational events become private audit arrays; verified snapshots protect database and private working data.

External participant • Process / activity • Data store • Cross-sheet reference → Data flow



- Audit data is stored as private daily JSON arrays, not an operational SQL audit table. Audit listing filters these files. Backup/restore permissions are separately checked.
- A backup contains a SQL dump plus application/private files, excluding sessions/cache and backup recursion. Its ZIP entries are AES-256 encrypted and the manifest is signed; imported archives are verified.
- Web restore reapplies the database and private working data after a pre-restore recovery snapshot. Application-code recovery is offline. Restore maintenance blocks ordinary requests; retention maintains managed archives and daily audit/application logs.
- D1-D4 are logical store groups; see the physical store catalog on sheet 8. A cross-sheet port refers to the numbered process on its named sheet.

Source evidence: application/core/MY_Controller.php:78-157 | application/libraries/Audit_file.php:5-52 | application/controllers/Audit.php:6-30 | application/controllers/Backups.php:6-155 | application/services/backup/backup_service.php:109-362 | application/services/backup/retention_service.php:21-135

Logical store catalog and scope

Physical implementation mapping for D1-D4; boxes define logical stores used on sheets 2-7.

Data store

D1 Identity and configuration
SQL: users, roles, user_roles, role_permissions, permissions,
login_attempts; areas, specifics, assets, locations, categories;
workflows, workflow_versions
Private session files: storage/sessions

D2 Document, file and custody records
SQL: hardcopy_documents, softcopy_documents, softcopy_revisions,
status_history, disposals, files, assignments, access_grants, transfers
Controlled file bytes: storage/documents/<random storage_name>
Softcopy assignments use assignments; hardcopy assignments update holder_id.

D3 Request and approval runtime
SQL: requests, workflow_steps, workflow_history, task_notification_receipts
Stored data: proposal payload, pinned workflow/version, active assignments,
decision history and per-user task occurrence receipts

D4 Audit and recovery files
Daily audit JSON arrays: storage/audit/YYYY-MM-DD.json
Encrypted AES-256 ZIP archives: storage/backups (or configured safe backup path)
Backups carry a signed manifest, SQL snapshot and application/private files.
Retention maintains managed backups, daily audit and application logs.

• Source-derived design snapshot of the Desktop checkout. Seed roles include Administrator, Document Control Officer, Plant Manager, Internal Auditor and Staff; permissions and workflow assignments remain configurable.
• No operational SQL table for audit is used. The notifications table exists in the schema, but current inbox/pending logic reads requests + workflow_steps + task_notification_receipts; notifications appears only in draft cleanup.
• revision_artifacts, settings and schema_migrations exist in the SQL snapshot; no active application read/write was found, so they are schema context rather than depicted business-flow stores.
• Office/WPS review uses optional local protocol handlers and a signed content URL. CDN assets/ngrok tooling are deployment/client dependencies, not document-approval data services.
• Backups contain a SQL dump plus application/private files, excluding transient sessions/cache and backup recursion. Web restore reapplies the database and private working data; application-code recovery is offline.

Source evidence: database/pk_dts.sql:30-1150 | database/migrations/20261013_user_role_assignments.sql:2-13 | application/config/config.php:21-24 | application/models/Permission_model.php:23-49 | application/models/Notification_model.php:89-225 | application/services/files/file_service.php:171-228 | application/libraries/Audit_file.php:5-52 | application/services/backup/backup_service.php:109-156,270-362